DECISION MAKING

56. Consider the following sum:

• + 1• + 2 • + • 3 + • 1 = 21 •

In the above sum, • stands for

(a) 4 (b) 5

(c) 6 (d) 8

Ans. (d) Suppose • equal to x then

$$\bullet + 1\bullet + 2\bullet + \bullet 3 + \bullet 1 = 21\bullet$$

$$x + (10 + x) + (20 + x) + (10x + 3) + (10x + 1) = 210 + x$$

$$23x + 34 = 210 + x$$

$$22x = 176$$

$$\boxed{x = 8}$$

57. Consider the following pattern of numbers:

8	10	15	13
6	5	7	4
4	6	8	8
6	11	16	?

What is the number at ? in the above pattern?

(a) 17 (b) 19

(c) 21 (d) 23

Ans. (a)

8	10	15	13
6	5	7	4
4	6	8	8
6	11	16	?

$\because$ $8 + 4 - 6 = 6$

$10 + 6 - 5 = 11$

$15 + 8 - 7 = 16$

$13 + 8 - 4 = \boxed{17}$

58. How many diagonals can be drawn by joining the vertices of an octagon?

(a) 20 (b) 24

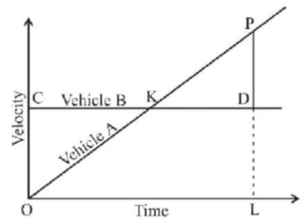
(c) 28 (d) 64

Ans. (a)

The number of diagonals $\dfrac{n(n-3)}{2}$

$$= \frac{8(8-3)}{2} = 4 \times 5 = \boxed{20}$$

59. The figure drawn below gives the velocity graphs of two vehicles A and B. The straight line OKP represents the velocity of vehicle A at any instant, whereas the horizontal straight line CKD represents the velocity of vehicle B at any instant. In the figure, D is the point where perpendicular from P meets the horizontal line CKD such that PD $= \frac{1}{2}$ LD :

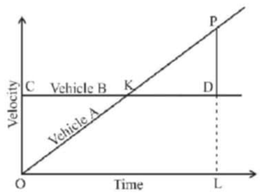


What is the ratio between the distances covered by vehicles A and B in the time interval OL?

(a) 1 : 2 (b) 2 : 3

(c) 3 : 4 (d) 1 : 1

Ans. (c)



$\because$ Velocity

$$v = \frac{d}{t}$$

$$\boxed{d = vt}$$

The distance covered by vehicle A = Area of ΔOPL

$$dA = \frac{1}{2} \times OL \times PL$$

$$= \frac{1}{2} \times OL \times (PD + DL)$$

$$= \frac{1}{2} \times OL \times \left(\frac{1}{2}LD + DL\right)$$

$$= \frac{1}{2} \times OL \times \frac{3}{2}LD$$

$$\left[\because PD = \frac{1}{2}LD\right]$$

$$\boxed{d_A \Rightarrow \frac{3}{4}OL \times L_D}$$

The distance covered by B = Area of figure OCDL

$d_B = OL \times L_D$

then,

$$\frac{d_A}{d_B} = \frac{\frac{3}{4} \times OL \times LD}{OL \times LD}$$

$$\frac{d_A}{d_B} = \frac{3}{2}$$

$$\boxed{d_A : d_B = 3 : 4}$$

60. A train 200 metres long is moving at the rate of 40 kmph. In how many seconds will it cross a man standing near the railway line?

(a) 12 (b) 15

(c) 16 (d) 18

Ans. (d)

$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$

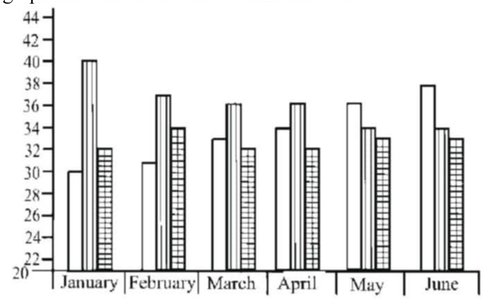
$$40 \times \frac{5}{18} = \frac{200}{t}$$

$$\frac{200}{18} = \frac{200}{t}$$

$$\boxed{t = 18 \text{ sec.}}$$

Directions for the following 3 (three) items :

The Following three items are based on the graph given below which shows imports of three different types of steel over a period of six months of a year. Study the graph and answer the three items that follow.



The figures in the brackets indicate the average cost per ton over six months period.

61. By how much (measured in thousands of tons) did the import of sheet steel exceed the import of coil steel in the first three months of the year?

(a) 11 (b) 15

(c) 19 (d) 23

Ans. (c) Inport of sheet steel in the first three months of the year = 40 + 37 + 36 = 113

Import of coil sheet in the first three months of the year = 30 + 31 + 33 = 94

Repuired difference = 113 – 94

$\Rightarrow$ 19

62. What was the approximate total value (in $) of sheet steel imported over the six months period?

(a) 45,555 (b) 50,555

(c) 55,550 (d) 65,750

Ans. (c)

Import of steel sheet ones the six momths period = 40 + 37 + 36 + 36 + 34 + 34 = 217 tons

$\because$ average cast of steel sheet = 256 $/ton

$\because$ Total cast = 217 × 256

= 55552

$\approx$ 55550 (approximate)

63. What was the approximate ratio of sheet steel and scrap steel imports in the first three months of the year?

(a) 1 : 1 (b) 1.2 : 1

(c) 1.4 : 1 (d) 1.6 : 1

Ans. (b) Import of sheet steel in the first three months = 40 + 37 + 36 = 113

Import of scrap steel in the first three months

$\Rightarrow$ 32 + 34 + 32

$\Rightarrow$ 98

$$\text{Required Ratio} = \frac{113}{98}$$

$$= \frac{1.15}{1} = \frac{1.2}{1} \text{ (approximate)}$$

$\Rightarrow$ 1.2 : 1

64. If X is between –3 and –1, and Y is between –1 and 1, then $X^2 - Y^2$ is in between which of the following?

(a) –9 and 1 (b) –9 and –1

(c) 0 and 8 (d) 0 and 9

Ans. (d)

$\because$ Range of x = –3 to – 1

Range of y = –1 to + 1

Max value of $x^2 = (-3)^2 = 9$

Min value of $x^2 = (-1)^2 = 1$

Max value of $y^2 = 1^2 = 1$

Min value of $y^2 = (0)^2 = 0$

then max value of $(x^2 - y^2) = 9 - 0$

min value of $(x^2 - y^2) = 1 - 1 = 0$

$$\boxed{\text{Range of } (x^2 - y^2) \text{ is } -0 \text{ to } 9}$$

65. X and Y are natural numbers other than 1, and Y is greater than X. Which of the following represents the largest number?

(a) XY (b) X / Y

(c) Y / X (d) (X + Y) / XY

Ans. (a) $\because$ x and y are natural numbers other than 1 and y is greater than x

$\because$ x = 2 and y = 3

by option,

(a) $xy = 2 \times 3 \Rightarrow 6$

(b) $\frac{x}{y} = \frac{2}{3}$

(c) $\frac{y}{x} = \frac{3}{2}$

(d) $\frac{x+y}{xy} = \frac{3+2}{2 \times 3} = \frac{5}{6}$

Hence option (a) "x y" represents the largest number.